

Data Cleaning Log

Project

Business Data Cleaning, Validation & Excel Reporting

Dataset

- Original File: `raw_orders.xlsx`
- Cleaned File: `cleaned_orders.xlsx`

Issues Identified

- Leading and trailing spaces in text fields.
- Inconsistent capitalization in customer names, regions, categories, and cities.
- Missing values in Region, Ship Mode, Discount, and Date fields.
- Invalid or inconsistent date formats.
- Ship dates occurring before order dates.
- Duplicate records and duplicate Order IDs.
- Invalid discount values (negative or greater than the allowed range).
- Sales and profit calculation mismatches.
- Inconsistent Order Status and Payment Status values.

Cleaning Actions Performed

- Trimmed extra spaces using Excel TRIM.
- Standardized text capitalization using PROPER where appropriate.
- Corrected inconsistent category and sub-category names.
- Standardized Region, State, City, Segment, Ship Mode, Payment Status, and Order Status values.
- Converted Order Date and Ship Date to a consistent date format.
- Calculated Shipping Delay Days.
- Identified and flagged duplicate Order IDs.
- Removed exact duplicate rows.
- Filled missing Region values with **Unknown**.
- Filled missing Ship Mode values with **Unknown**.
- Replaced missing Discount values with **0** only when other sales information was valid.
- Flagged negative discounts.
- Flagged discounts above the acceptable range.
- Flagged records where Ship Date was earlier than Order Date.
- Created calculated columns for Sales, Profit, Profit Margin, Order Month, Order Year, and Data Quality Flag.

Business Rules Applied

- Missing Region → Unknown
- Missing Ship Mode → Unknown
- Missing Discount → 0 (only if valid)
- Negative Discount → Invalid

- Excessive Discount → Invalid
- Cancelled Orders excluded from completed sales summary
- Failed Payments excluded from completed sales summary
- Refunded Orders summarized separately
- Ship Date before Order Date flagged as Invalid

Records Removed

- Exact duplicate rows were removed where applicable.

Records Flagged

- Duplicate Order IDs with conflicting information
- Invalid Discount values
- Invalid Shipping Dates
- Missing mandatory fields

Assumptions

- Missing discounts were treated as zero only when all required sales fields were available.
- Profit Margin was calculated as $\text{Profit} \div \text{Sales}$.
- Invalid records were retained but flagged for business review.

Limitations

- Some conflicting duplicate records require manual verification.
- Business assumptions were based on the rules provided in the assignment.